

Università Bocconi

Macroeconomics

(Economics – Module 2 -30066)

Prof. Edoardo Teso

Chapter 21:

Exchange Rate Regimes

The Medium Run

- As we saw before, a fixed exchange rate has drawbacks in the short run:
 - Losing the possibility of having a depreciation
 - Losing the possibility of lowering the interest rate
- Then, why would a country chose a fixed exchange rate regime?
- The answer has to do with what happens in the medium run...
- Remember that the real exchange rate is: $\varepsilon = \frac{EP}{P^*}$

which can be adjusted either:

- **through a change in E .** If the domestic price level and foreign price level do not change in the short run, this is the **only way to adjust the real exchange rate in the short run.**
- **through a change in the domestic price level relative to the foreign price level.** In the medium run, as prices adjust, this option is **open even to a country with a fixed nominal exchange rate.**

The Medium Run

How do we write the *IS* relation in an open economy with fixed exchange rates?

- Remember that in the open economy, goods market equilibrium is:

$$Y = \underset{(+)}{C(Y - T)} + \underset{(+, -)}{I(Y, r)} + G + \underset{(-, +, -)}{NX(Y, Y^*, \varepsilon)}$$

- Remember the definition of the real interest rate:

$$r = i - \pi^e$$

- Remember that the real exchange rate is:

$$\varepsilon = \frac{EP}{P^*}$$

- And remember that with fixed exchange rates: $E = \bar{E}$
- With perfect capital mobility, interest parity implies that: $i = i^*$

The Medium Run

Putting all together: the *IS* relation in an open economy with fixed exchange rates is given by:

$$Y = C(Y - T) + I(Y, i^* - \pi^e) + G + NX\left(Y, Y^*, \frac{\bar{E}P}{P^*}\right)$$

Which can be re-written in more compact form as:

$$Y = Y\left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^*\right)$$

Demand (and so output) depends:

- negatively on the real exchange rate
- positively on government spending and negatively on taxes
- negatively on the domestic real interest rate
- positively on foreign output through the effect on exports

The Medium Run

- Assume that a country is stuck with a too high real exchange rate: recession + trade deficit.
- If fixed exchange rates, no way to reduce real exchange rate in the short run. And in the medium run?

- The Phillips curve relation in the medium run:

$$\pi - \pi^e = (\alpha/L)(Y - Y_n)$$

- Assume constant expected inflation so that :

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n)$$

- Assuming that the other country is operating at potential (so inflation is equal to expected inflation), then if a country is in a recession, its **inflation will be lower than in the other country.**

The Medium Run

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n)$$

- If inflation is below the inflation in the other country, this implies that the real exchange rate depreciates as time passes (in the medium run).
- In the short run, a fixed nominal exchange rate implies a fixed real exchange rate.
- In the medium run, the real exchange rate can adjust even if the nominal exchange rate is fixed.
- This adjustment is achieved through movements in the relative price levels over time.
- The process of adjustment can be long and painful with output below potential for long time. Less painful ways?
 - one-time devaluation -> immediate real depreciation

Exchange Rate Crises under Fixed Exchange Rates

- Suppose that people believe that there will be soon a devaluation (or, even more extreme, a shift to flexible exchange rates). Why?
 - large CA deficit and recession
 - High inflation, leading to real appreciation (and so steady worsening of trade balance)
 - or simply because markets think so (self-fulfilling prophecy!)

Given the (approximate version of the) interest parity condition:

$$i_t = i_t^* - \frac{(E_{t+1}^e - E_t)}{E_t}$$

- With fixed exchange rates, then the second term is zero, and domestic and foreign interest rates are equal
- If markets expect a devaluation, the the second term is negative, and to maintain parity now the domestic interest rate must be higher than the foreign one

Exchange Rate Crises under Fixed Exchange Rates

$$i_t = i_t^* - \frac{(E_{t+1}^e - E_t)}{E_t}$$

For example, assume markets have the following expectations:

- With a 50% chance, there will be a devaluation of 20%
- With a 50% chance, no devaluation.

Then, the second term is equal to $0.5 \times 0 + 0.5 \times (-0.2) = -0.1 = -10\%$

In light of this, if the Central Bank wants to maintain parity, it must increase the domestic interest rate by 10%!

Exchange Rate Crises under Fixed Exchange Rates

Faced with the expectations of devaluation, the government has three options:

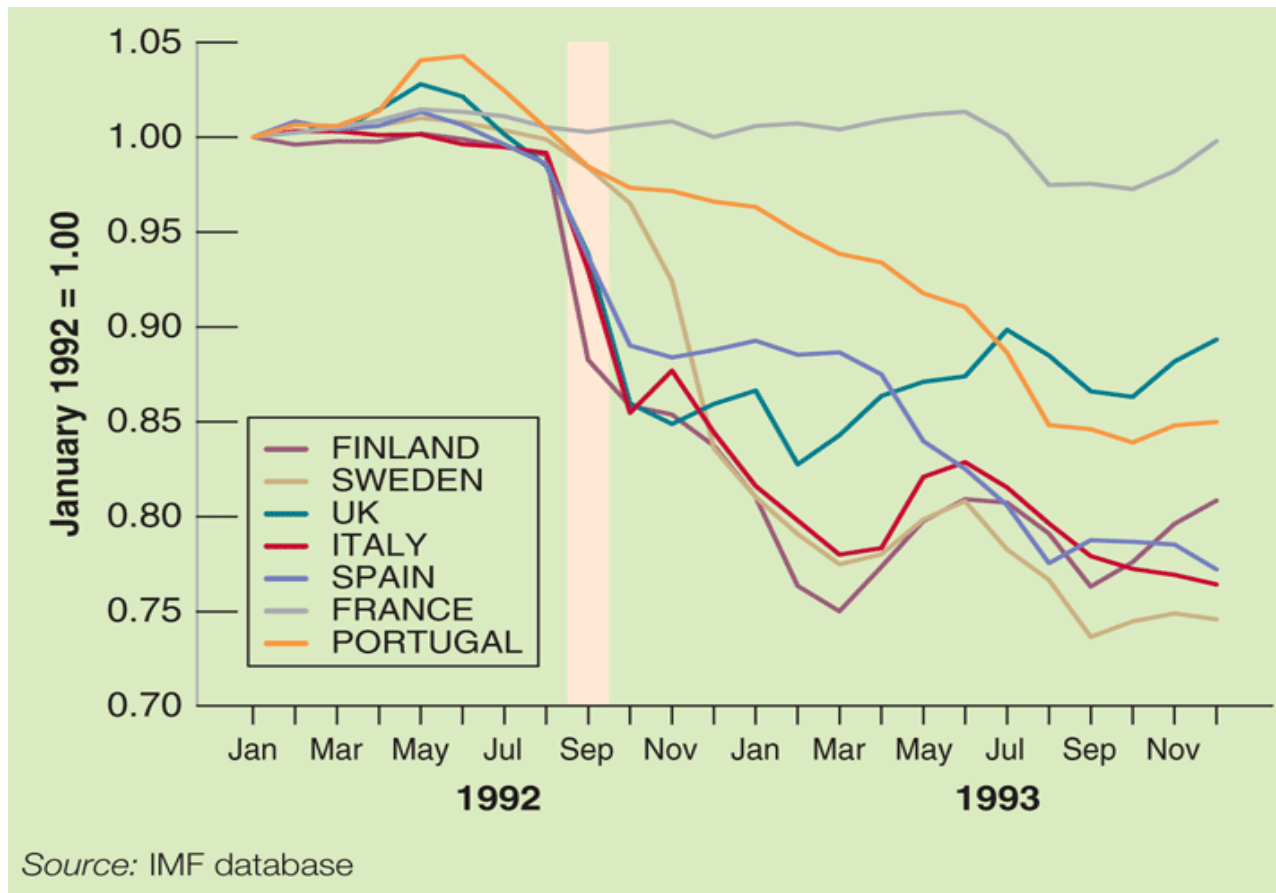
1. Try to convince the markets that no devaluation is coming. Of course, this might be hard to do...
2. Give in and devalue.
3. Fight and maintain the parity, at the cost of high interest rates and a potential recession.
 - To do so, what happens is:
 - since investors, with current expectations, are currently selling domestic currency to buy foreign currency (and invest that in foreign bonds), the exchange rate is decreasing.
 - To overcome this, the Central Bank must sell part of their reserves in foreign currency and buy domestic currency.
 - This prevents the exchange rate from falling, increasing interest rate.
 - More likely to fight and maintain the parity if expected devaluation is small

FOCUS: The 1992 EMS Crisis

- Remember (from previous chapter) the effect on interest rates of German reunification + EMS' narrow bands.
- Countries in the EMS forced to raise their interest rates to maintain parity with Germany, leading to painful consequences for the economy
- In September 1992, financial markets became convinced devaluations were coming soon.
- Speculative attacks on currencies of many countries
- Response:
 - First, press conferences announcing no devaluation was coming
 - Second, further interest rates increases to maintain parity at all costs
 - Third, many countries devalued (Spain) or exited the EMS (Italy, UK)

FOCUS: The 1992 EMS Crisis

Exchange Rates of Selected European Countries Relative to the Deutsche Mark,
January 1992 to December 1993



Exchange rate movements under flexible exchange rates

- In reality, exchange rates do not move one-to-one with nominal interest rates...reality is more complicated...
- Recall the interest parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

- Multiply both sides by E_{t+1}^e gives:

$$E_t = \frac{1+i_t}{1+i_t^*} E_{t+1}^e$$

- In the previous chapter, we assumed that E_{t+1}^e was constant.
- In reality, the expected E one year from now depends on expectations of future interest rates and the future exchange rate. We can re-write E_{t+1}^e as:

$$E_{t+1} = \frac{1+i_{t+1}}{1+i_{t+1}^*} E_{t+2}^e$$

Exchange rate movements under flexible exchange rates

- Continue to solve forward in time for the expected exchange rate:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \cdots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \cdots (1 + i_{t+n}^{*e})} E_{t+n+1}^e$$

- Take for instance $n=10$ years. Then, the exchange rate depends on:
 - Current and expected domestic and foreign interest rates for each year over the next n (in this case 10) years.
 - The expected exchange rate n (in this case 10) years from now.

Exchange rate movements under flexible exchange rates

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \cdots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \cdots (1 + i_{t+n}^{*e})} E_{t+n+1}^e$$

Implications of this equation:

- Today's exchange rate will move one-for-one with the future expected exchange rate.
- Today's exchange rate will move when future expected interest rates move in either country.
- Because today's exchange rate moves with any change in expectations, the **exchange rate will be volatile** (it will move frequently and perhaps by large amounts).



A country that decides to operate under flexible exchange rates must accept the fact that it will be exposed to substantial exchange rate fluctuations over time.

Exchange rate movements under flexible exchange rates

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \cdots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \cdots (1 + i_{t+n}^{*e})} E_{t+n+1}^e$$

Major sources of fluctuations

- News on current account : since investors expect that eventually current account will be balanced (you cannot borrow or lend forever), any news of changes in current account will lead investors to expect movements in future E (which will be needed to restore balance)
- News of changes in monetary policy (especially if perceived to be persistent rather than temporary).

Note: what moves E_t are news, not changes that were already expected.

Choosing between exchange rates regimes

- So far, everything seems to justify the choice of flexible exchange rates:
 - short-run flexibility (even if in the medium run real exchange rate adjusts)
 - no risk of an exchange rate crisis....
- Only caveat: possible wild variations in the exchange rate.
- 2 exceptions:
 - What if countries are tightly integrated and trade a lot?
 - What if the Central Bank in a country cannot be trusted to follow a responsible monetary policy?

Common currency areas

What if countries are tightly integrated and trade a lot?

- They could adopt a common currency (extreme version of fixed exchange rates among a group of countries).
- Big advantages: lower transaction costs and ease to compare prices among countries (great for competition).
- According to Robert Mundell, an **optimal currency area** needs to satisfy one of the three conditions:
 - The countries have to experience similar shocks
 - In this case, they would have followed similar monetary policies anyway even with flexible exchange rates
 - If the countries experience different shocks, they must have flexible prices and wages
 - In this case, they could lower prices and wages to stay competitive, if needed.
 - If the countries experience different shocks, they must have high factor mobility.
 - In this case, people can move across countries to adjust to shocks

Common currency areas

- Are the **United States** an optimal currency area?
 - Different states are affected by different shocks
 - But high factor mobility (workers willing to move across states)
- There is less agreement on whether **Europe** constitutes an optimal currency area.
 - Different shocks across countries
 - Prices and wages adjust slowly
 - Little factor mobility
- Southern Europe suffering from low demand lost the ability to benefit from lower exchange rates relative to their commercial partners
- (Note: benefits of the Euro are clear: for instance elimination of foreign exchange transactions in the euro area led to a reduction in transaction costs of 0.5% of the combined GDP of these countries.)

Limiting Central Bank's Ability to Conduct Monetary Policy

- What if the CB in a country **cannot be trusted to follow a responsible monetary policy?** Typical in cases of history of high inflation
- Possible solution: tie your hands with a fixed exchange rate!
- Caveat: you need to convince investors that the exchange rate will remain fixed also in the future.
- Possible ways to do so:
 - Make the fixed exchange rate be part of a more general macroeconomic package.
 - Make a **hard peg** so that it is symbolically or technically harder to change the parity.
 - **Dollarization:** Replace the domestic currency with the dollar (today, for instance: Ecuador, Zimbabwe, El Salvador, Panama).
 - **Currency board:** A central bank stands ready to exchange foreign currency for domestic currency at the official exchange rate set by the government + they cannot buy and sell government bonds

FOCUS: Lessons from Argentina's currency board

- In 1991, Argentina's president announced the adoption of a currency board that would stand ready to exchange its pesos for dollars on demand (after years of massive inflation)
- The currency board appeared to work well for a while as inflation fell from 2300% in 1990 to 4% in 1994.
- However, the appreciation of the dollar in the second half of the 1990s led to a decrease in demand for Argentina's goods, leading to declines in output.
 - Recession led investors to worry about a possible government default, so they started asking for even higher interest rates
 - Investors started to expect a future return to flexible exchange rates, further driving up interest rates.
- In early 2002, Argentina gave up the currency board and let the peso float.